Classical Field Theory — Exercise Sheet 26

Scattering evolution and soliton shifts

Solutions

Use the standing real KdV solution $u \in C^\infty(\mathbb{R}; \mathcal{S})$ of Lecture 24 and the Jost/scattering conventions of Lecture 25. The first problem fills the analytic step left sketched in Lecture 26. The later finite-rank construction prescribes its norming coefficients; it makes no additional theorem about birth of eigenvalues.

Exercise 26.1 (Time differentiation at real energy). Fix real $k \neq 0$. For the right Jost solution put $f_+(t,x,k) = e^{ikx} m(t,x,k)$, with

$$m = 1 + \int_x^\infty \frac{e^{2ik(y-x)} - 1}{2ik}\, u(t,y) m(t,y,k)\, \mathrm{d}y.$$

(a) Prove that this Volterra construction is continuously differentiable in t. On every compact time interval prove $m_t \to 0$ as $x \to +\infty$, together with the spatial derivative limits needed to compare both the function and derivative tails of $(f_+)_t - B f_+$. Here $B = -4\partial_x^3 + 6u\partial_x + 3u_x$.

(b) Prove the right and left temporal equations, justify differentiability of a, b from finite-point Wronskians, and prove $r(k,t) = r(k,t_0) e^{8ik^3(t-t_0)}$. Do not pair the oscillatory Jost solutions in L^2.

Solution. Fix a compact time interval J and a finite R. The real-energy kernel is bounded by $|k|^{-1}$. Define the tails

$$A_J(x) = |k|^{-1} \int_x^\infty \sup_{t\in J} |u(t,y)|\, \mathrm{d}y,$$
$$B_J(x) = |k|^{-1} \int_x^\infty \sup_{t\in J} |u_t(t,y)|\, \mathrm{d}y.$$

They are finite and tend to zero at the right end. The nth ordered iterate has bound $A_J(R)^n/n!$ on $x \geq R$. Differentiating one of its n coefficient factors and summing over all positions gives the bound $B_J(R) A_J(R)^{n-1}/(n-1)!$. Indeed, the time-differentiated point can occupy any position among the ordered remaining $n-1$ points. These scalar series converge uniformly. Termwise time differentiation is therefore justified, gives continuous m_t, and the refined tail bound is

$$|m_t(t,x,k)| \leq B_J(x) e^{A_J(x)} \longrightarrow 0 \quad \text{uniformly for } t \in J.$$

The same series gives $|m| \leq e^{A_J(x)}$ and $m \to 1$ uniformly there.

Differentiating in space, including the lower endpoint, gives

$$m_x = -\int_x^\infty e^{2ik(y-x)} u m\, \mathrm{d}y, \qquad m_{xx} = um - 2ikm_x.$$

The first expression tends uniformly to zero. For every $j \geq 0$, further differentiation obeys $m^{(j+2)} = D_x^j(um) - 2ikm^{(j+1)}$. Induction gives bounded spatial derivatives and their zero limits for all positive derivative orders, uniformly on J. Time differentiation of the integral for m_x gives

$$(m_t)_x = -\int_x^\infty e^{2ik(y-x)} (u_t m + u m_t)\, \mathrm{d}y.$$

The already proved uniform bounds and integrable tails justify it and make it tend to zero. Differentiating m_{xx} in time also gives $(m_t)_{xx} = u_t m + u m_t - 2ik(m_t)_x$, with zero right limit. These identities establish the local regularity needed below.

Thus, through every required spatial order, $e^{-ikx}\partial_x^j f_+ \to (ik)^j$. Also $e^{-ikx}(f_+)_t \to 0$ and $e^{-ikx}\partial_x(f_+)_t \to 0$. The limits through order four show that Bf_+ and its derivative have the same tails as $4ik^3 e^{ikx}$ and its derivative. Order five and the mixed second derivative justify applying L to $(f_+)_t - Bf_+$. Differentiating $Lf_+ = k^2 f_+$ and using the pointwise Lax formula gives

$$(L - k^2)\big((f_+)_t - Bf_+\big) = 0.$$

Hence $z = (f_+)_t - Bf_+ + 4ik^3 f_+$ solves that equation and has both zero right Jost tails. The solution $f_+ + z$ has exactly the same two normalisations as f_+, so Lecture 25's uniqueness forces $z = 0$. We have proved $(f_+)_t = Bf_+ - 4ik^3 f_+$.

The reflected Volterra estimates give the corresponding regularity at the left end. Apply the original Lax equation to that left solution; reflection is used for estimates, not to assert that $u(t,-x)$ solves the same KdV equation. Since $Be^{-ikx} \sim -4ik^3 e^{-ikx}$ there, uniqueness gives $(f_-)_t = Bf_- + 4ik^3 f_-$.

At a fixed finite spatial point the Wronskian formulas are

$$a = \frac{W(f_-, f_+)}{2ik}, \qquad b = \frac{W(f_+(\cdot,-k), f_-)}{2ik}.$$

The just-proved C^1 time dependence of functions and first derivatives makes both coefficients C^1. Differentiate $f_- = af_+(-k) + bf_+(k)$. Use the two temporal equations, including the reversed sign at $-k$, and cancel the B terms. The result is

$$a_t f_+(-k) + (b_t - 8ik^3 b) f_+(k) = 0.$$

Their nonzero Wronskian $2ik$ gives $a_t = 0$ and $b_t = 8ik^3 b$. Since $a \neq 0$, division and integration give $r(k,t) = r(k,t_0)e^{8ik^3(t-t_0)}$. No integral of a squared Jost function was used.

Exercise 26.2 (A positive finite-rank inverse equation). Take distinct $\kappa_j > 0$ and $c_j > 0$, $1 \le j \le N$, with $F(s) = \sum_j c_j e^{-\kappa_j s}$. Consider the right Marchenko equation

$$K(x,y) + F(x+y) + \int_x^\infty K(x,z)F(z+y)\,\mathrm{d}z = 0, \qquad y \ge x.$$

The finite positive reflectionless case of the inverse-scattering theorem is a *given*: these data are admissible and reconstruct a real rapidly decaying potential by $u = -2\,\mathrm{d}K(x,x)/\mathrm{d}x$. It is the zero-background specialization of the cited theorem in Lectures 25–26: Egorova–Grunert–Teschl, *Nonlinearity* **22** (2009), §4, Theorems 4.3–4.4, equations (4.26)–(4.30). Their $(\gamma_j^+)^2$ is c_j.

(a) Reduce the equation to a real symmetric $N \times N$ linear system. Prove that its matrix is positive definite and invertible for every real x.

(b) With $v_j(x) = \sqrt{c_j}e^{-\kappa_j x}$, define $\mathsf{C}_{jl} = v_j v_l/(\kappa_j + \kappa_l)$ and $\tau = \det(I + \mathsf{C})$. Prove $K(x,x) = (\log\tau)'$ and $u = -2(\log\tau)''$. Explain why no real pole arises from these positive data.

Solution. The integral equation itself makes $K(x,y)$ a linear combination of the N exponentials in y. Write $K = -\sum_j z_j e^{-\kappa_j y}$. Since the exponents are distinct, their coefficients give

$$z_j + c_j \sum_l \frac{e^{-(\kappa_j+\kappa_l)x}}{\kappa_j + \kappa_l} z_l = c_j e^{-\kappa_j x}.$$

After setting $q_j = z_j/\sqrt{c_j}$ and $v_j = \sqrt{c_j}e^{-\kappa_j x}$ this is

$$(I + \mathsf{C})q = v, \qquad \mathsf{C}_{jl} = \frac{v_j v_l}{\kappa_j + \kappa_l}.$$

For any real column a,

$$a^{\mathsf{T}}\mathsf{C}a = \int_x^\infty \left(\sum_j a_j\sqrt{c_j}e^{-\kappa_j z}\right)^2 \mathrm{d}z \geq 0.$$

Thus $a^{\mathsf{T}}(I + \mathsf{C})a \geq |a|^2$, proving invertibility and a strictly positive determinant at every finite x.

On the diagonal the kernel is $K(x,x) = -v^{\mathsf{T}}q = -v^{\mathsf{T}}(I + \mathsf{C})^{-1}v$. Differentiating each entry gives $\mathsf{C}' = -vv^{\mathsf{T}}$. Jacobi's finite determinant formula therefore yields

$$(\log\tau)' = \operatorname{tr}[(I + \mathsf{C})^{-1}\mathsf{C}'] = -v^{\mathsf{T}}(I + \mathsf{C})^{-1}v = K(x,x).$$

The supplied reconstruction formula gives $u = -2(\log\tau)''$. The entries are smooth and $\tau > 0$, so this expression and the kernel have no real poles. These finite-dimensional calculations solve the integral equation; the interpretation as complete scattering data remains the stated inverse-theorem input.

Exercise 26.3 (Two pulses and their shifts). Take $\kappa_1 > \kappa_2 > 0$ and prescribe $c_j(t) = c_j(0)e^{8\kappa_j^3 t}$, with $c_j(0) > 0$. Put

$$x_{j,0} = \frac{1}{2\kappa_j}\log\frac{c_j(0)}{2\kappa_j}, \qquad \eta_j = -2\kappa_j(x - 4\kappa_j^2 t - x_{j,0}).$$

(a) Evaluate the two-state determinant and find its interaction coefficient d_{12} in $\tau = 1 + e^{\eta_1} + e^{\eta_2} + d_{12}e^{\eta_1+\eta_2}$.

(b) Verify directly that $u = -2\partial_x^2\log\tau$ solves KdV. One may use, after deriving it by differentiating $\log\tau$,

$$\tau\tau_{xt} - \tau_x\tau_t + \tau\tau_{xxxx} - 4\tau_x\tau_{xxx} + 3\tau_{xx}^2 = 0$$

as a sufficient condition.

(c) Find the incoming and outgoing centre lines of each separated pulse and its net shift. Justify the two spatial derivatives in each moving-frame limit, rather than compare only pointwise values of τ.

The parameters $x_{j,0}$ describe the norming coefficients. They need not be actual minima positions during the interaction.

Solution. The two diagonal entries of C are e^{η_j}, while $\mathsf{C}_{12}^2 = 4\kappa_1\kappa_2 e^{\eta_1+\eta_2}/(\kappa_1+\kappa_2)^2$. Thus

$$\tau = 1 + e^{\eta_1} + e^{\eta_2} + \left(1 - \frac{4\kappa_1\kappa_2}{(\kappa_1+\kappa_2)^2}\right)e^{\eta_1+\eta_2}.$$

The coefficient is $d_{12} = (\kappa_1 - \kappa_2)^2/(\kappa_1+\kappa_2)^2$, strictly between zero and one.

For the PDE check set $q = \log\tau$. Dividing the proposed bilinear expression by τ^2 and differentiating the logarithm gives $q_{xt} + q_{xxxx} + 6q_{xx}^2$. Its x derivative, multiplied by -2, is $u_t - 6uu_x + u_{xxx}$ for $u = -2q_{xx}$.

To verify that expression, let $\zeta_j = -2\kappa_j$. Then $\partial_x \eta_j = \zeta_j$ and $\partial_t \eta_j = -\zeta_j^3$. In the symmetric exponential-pair expansion, constant/self pairs vanish. Pairs differing by only one η_j vanish by this dispersion relation. The remaining coefficient of $e^{\eta_1+\eta_2}$ is

$$3\zeta_1\zeta_2\left[d_{12}(\zeta_1+\zeta_2)^2 - (\zeta_1-\zeta_2)^2\right] = 0.$$

The chosen d_{12} proves its vanishing, hence the KdV equation.

In the faster frame $x = 4\kappa_1^2 t + X$, $\eta_2 \to +\infty$ as $t \to -\infty$ and $\eta_2 \to -\infty$ as $t \to +\infty$. The relevant determinants are consequently $e^{\eta_2}(1 + d_{12}e^{\eta_1})$ incoming and $1 + e^{\eta_1}$ outgoing. The incoming centre solves $d_{12}e^{\eta_1} = 1$, giving

$$x_1^{\text{in}} = 4\kappa_1^2 t + x_{1,0} + \frac{\log d_{12}}{2\kappa_1}, \qquad x_1^{\text{out}} = 4\kappa_1^2 t + x_{1,0}.$$

In the slower frame the roles are reversed: $\eta_1 \to -\infty$ incoming and $\eta_1 \to +\infty$ outgoing. Thus

$$x_2^{\text{in}} = 4\kappa_2^2 t + x_{2,0}, \qquad x_2^{\text{out}} = 4\kappa_2^2 t + x_{2,0} + \frac{\log d_{12}}{2\kappa_2}.$$

Subtracting the two intercepts gives $\Delta x_1 = -\log d_{12}/(2\kappa_1) > 0$ and $\Delta x_2 = \log d_{12}/(2\kappa_2) < 0$.

On each compact set of X, the discarded exponential has an exponentially small factor in $|t|$. Its first two X derivatives have the same factor, since each exponent is affine in X. The retained factors $1+e^{\eta_j}$ and $1+d_{12}e^{\eta_j}$ are bounded below by one. The same estimates therefore hold after two derivatives of their logarithms. They give separated profiles $-2\kappa_j^2 \operatorname{sech}^2$ with the centre lines just found, not merely a pointwise determinant comparison.